

①

A.) Derivation for linear a_i

$$g_i(x) = a_i(x - x_i) + b_i$$

$$y_i = g_i(x_i) = a_i(x_i - x_i) + b_i$$

$$\rightarrow y_i = b_i$$

$$y_{i+1} = g_i(x_{i+1}) = a_i(x_{i+1} - x_i) + y_i$$

$$y_{i+1} - y_i = a_i(x_{i+1} - x_i)$$

$$a_i = \frac{y_{i+1} - y_i}{x_{i+1} - x_i}$$

B.) Eqn. (3) and (4) cubic

$$(3) \quad 3a_i h_i^2 + 2b_i h_i + c_i = c_{i+1}$$

$$(4) \quad 6a_i h_i + 2b_i = 2b_{i+1}$$

General eqn

$$g_i(x) = a_i(x - x_i)^3 + b_i(x - x_i)^2 + c_i(x - x_i) + d_i$$

Constraints

Define

$$\begin{cases} g_i(x_i) = y_i & h_i = x_{i+1} - x_i \\ g_i(x_{i+1}) = y_{i+1} & \eta_i = y_{i+1} - y_i \\ g_i'(x_{i+1}) = g_{i+1}'(x_{i+1}) \\ g_i''(x_{i+1}) = g_{i+1}''(x_{i+1}) \end{cases}$$

Start w/ derivative $g'(x) = dg/dx$

$$g_i'(x) = 3a_i(x - x_i)^2 + 2b_i(x - x_i) + c_i$$

$$= g_{i+1}'(x) = 3a_{i+1}(x - x_{i+1})^2 + 2b_{i+1}(x - x_{i+1}) + c_{i+1}$$

Plug in $x = x_{i+1}$

$$3a_i h^2 + 2b_i h_i + c_i = 0 + 0 + c_{i+1}$$

$$3a_i h^2 + 2b_i h_i + c_i = c_{i+1}$$

Next $g''(x)$

$$g_i''(x) = 6a_i(x - x_i) + 2b_i$$

$$= g_{i+1}''(x) = 6a_{i+1}(x - x_{i+1}) + 2b_{i+1}$$

Plug in $x = x_{i+1}$

$$6a_i h_i + 2b_i = 0 + 2b_{i+1}$$

$$6a_i h_i + 2b_i = 2b_{i+1}$$

$$(1.) \quad (3) \quad 3a_i h_i + b_i = b_{i+1} \quad \text{Solve for } a_i$$

$$(2) \quad a_i h_i^3 + b_i h_i^2 + c_i h_i = \tau_i \quad \text{Solve for } c_i$$

$$(4) \quad 3a_i h_i^2 + 2b_i h_i + c_i = c_{i+1}$$

$$\Rightarrow \frac{1}{3} h_i b_i + \frac{2}{3} (h_i + h_{i+1}) b_{i+1} + \frac{1}{3} h_{i+1} b_{i+2} = \frac{\tau_{i+1}}{h_{i+1}} - \frac{\tau_i}{h_i}$$

$$(3) \Rightarrow a_i = \frac{b_{i+1} - b_i}{3 h_i}$$

$$(2) \Rightarrow \left(\frac{b_{i+1} - b_i}{3 h_i} \right) h_i^3 + b_i h_i^2 + c_i h_i = \tau_i$$

$$c_i = \frac{\tau_i}{h_i} - \frac{1}{3} (b_{i+1} - b_i) h_i - b_i h_i$$

$$c_{i+1} = \frac{\tau_{i+1}}{h_{i+1}} - \frac{1}{3} (b_{i+2} - b_{i+1}) h_{i+1} - b_{i+1} h_{i+1}$$

Plug into (4)

$$\begin{aligned} & 3 \left(\frac{b_{i+1} - b_i}{3 h_i} \right) h_i^2 + 2 b_i h_i + \frac{\tau_i}{h_i} - \frac{1}{3} (b_{i+1} - b_i) h_i - b_i h_i \\ &= \frac{\tau_{i+1}}{h_{i+1}} - \frac{1}{3} (b_{i+2} - b_{i+1}) h_{i+1} - b_{i+1} h_{i+1} \end{aligned}$$

Move τ to right, bring terms to left and combine

$$\frac{2}{3} (b_{i+1} - b_i) h_i + b_i h_i + \frac{1}{3} (b_{i+2} - b_{i+1}) h_{i+1} + b_{i+1} h_{i+1}$$

$$= \frac{\tau_{i+1}}{h_{i+1}} - \frac{\tau_i}{h_i}$$

$$\begin{aligned} & \frac{2}{3} b_{i+1} h_i - \frac{2}{3} b_i h_i + b_i h_i + \frac{1}{3} b_{i+2} h_{i+1} - \frac{1}{3} b_{i+1} h_{i+1} \\ & + b_{i+1} h_{i+1} = \frac{\tau_{i+1}}{h_{i+1}} - \frac{\tau_i}{h_i} \end{aligned}$$

Combine + pull in b_{i+1} terms

$$\frac{1}{3} h_i b_i + \frac{2}{3} (h_i + h_{i+1}) b_{i+1} + \frac{1}{3} h_{i+1} b_{i+2} = \frac{\tau_{i+1}}{h_{i+1}} - \frac{\tau_i}{h_i}$$